Bargaining over Leasing Contracts

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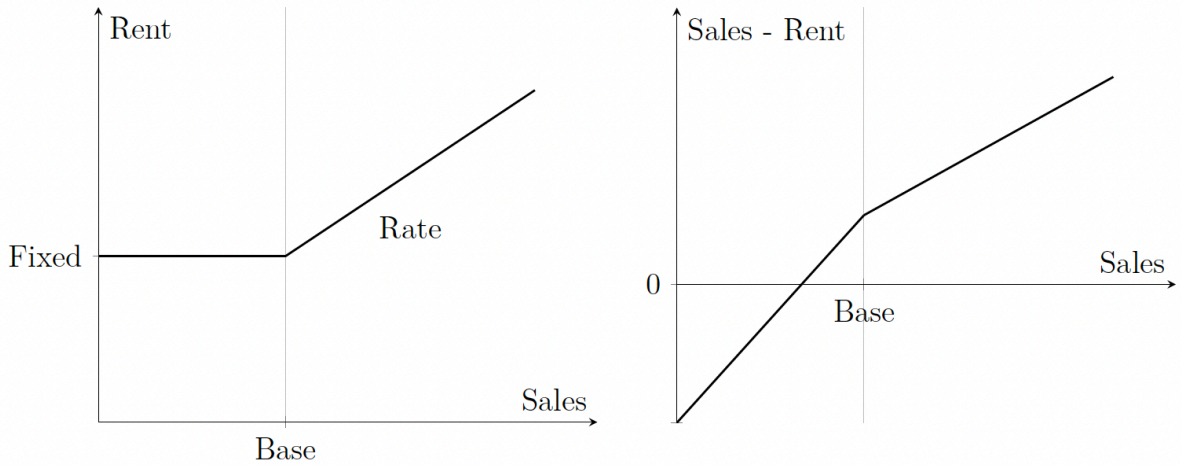
Question: What Determines Leasing Contracts?

- **Tenant Leasing in Shopping Malls**:
  - Fixed + Commission with a *kinked* structure



(a) Share of Shopping Mall (b) Share of Tenant

Figure 1. How Sales is Shared with Shopping Mall and Tenant

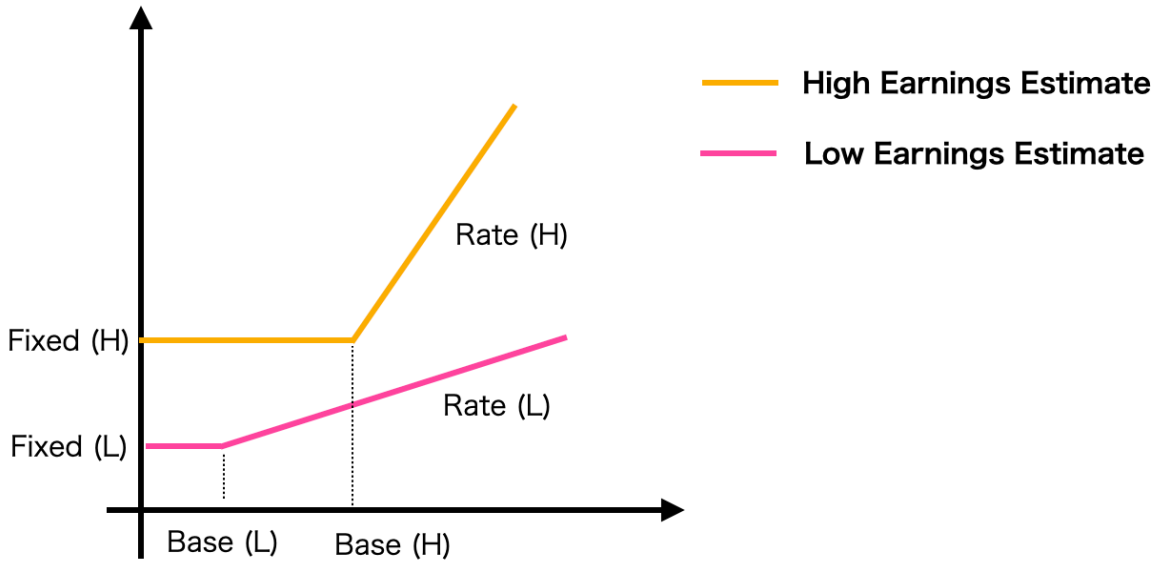
- Key Questions
  - What factors determine the contract structure?
  - To what extent do these factors matter?
  - How would monthly rent change under counterfactual scenarios?

Academic Interest: Understanding B2B Transactions

- B2B transactions involve bargaining over transfer schemes, i.e., *contracts*
  - Rarely under perfect competition
  - Auctions or other open mechanisms are rarely used
  - Limited empirical understanding of non-linear pricing
- Theory: Kihlstrom and Roth (1982) on insurance contract bargaining
  $\rightarrow$ **Risk aversion** matters, but other factors arise empirically
- Bargaining outcome shaped by three elements:
  - Bargaining frontier: Earnings estimate, party preferences
  - Break-up point: Availability of outside options
  - Bargaining power: Risk aversion (Rubinstein 1982, 1985), past performance and action (Joskow 1987), and others
- Study how these factors are empirically determined and affect contract forms

Result Summary 1: Empirical Contract Pattern

- Risk aversion affects the contract form in two ways:
  - Risk aversion decreases commission rate
  - The more risk averse mall more likely responds to variance of sales
- The lower earnings estimate increases the prob. of commission
  - Even with lower fixed component and lower rate of commission



- Empirical patterns of break-up points and bargaining power

Practical Interest: Recent Regulatory Focus

- **Regulatory shift**: focus on *fairness in individual transactions*, not just on preserving market-wide competition
  - Drivers: protecting SMEs and the rise of digital platforms
  - Examples: Germany's §20 GWB, Japan's JFTC Guidelines on Abuse of Superior Bargaining Position
- **Empirical challenge**: identifying unfairness in complex contracts
  - Overtly abusive clauses have largely disappeared, complicating detection
- **Example**: Singapore's Fair Tenancy Industry Code of Conduct
  - Standards for balanced lease terms which prevent landlords from
    - using multiple rent formulas
    - imposing one-sided termination rights
- Study how power balance appears in the contract form

Result Summary 2: Simulation of Fairer Bargaining

- Endogeneity concerns
  - Risk aversion of shopping mall manager
    - Manager becomes more risk averse when mall stands in stronger position
    - This correlation is confirmed in the estimation
  - Earnings estimate for the contract period
    - Position affects not only the contract form but also the sales forecast
    - Anecdote: Optimistic tenant vs Pessimistic mall
- Simulation result: fairer bargaining does not always reduce rent
  - Total rent does not necessarily decrease
  - Commission component rises sharply with risk adjustments
  - A risk-loving contract can triple rent under fairer terms
- Contract form or amount of rent are not always evidence of unfairness

Related Literature

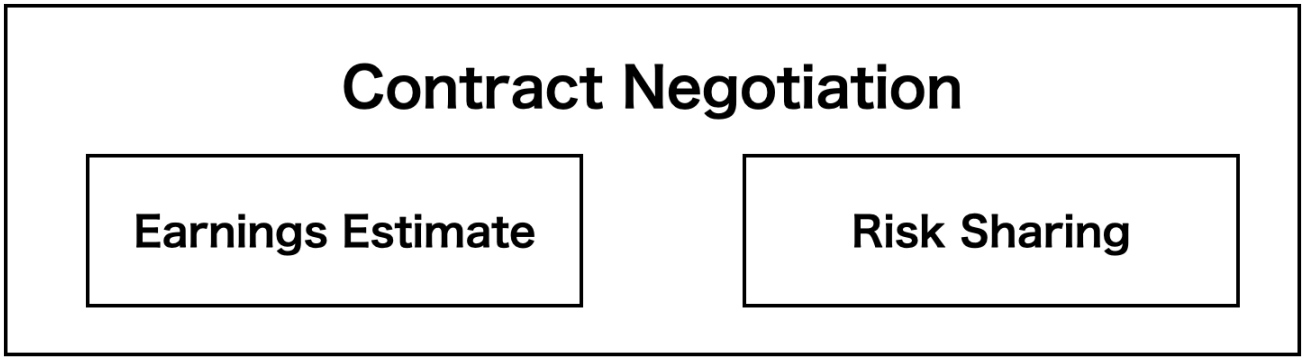
- Empirical bargaining (Lee et al. 2020)
  - **Cooperative approach**: Separating bargaining problem from power balance makes model tractable
  - **Contribution: Bargaining over unobserved objects**
- Sources of bargaining powers (Rubinstein 1982, Joskow 1987, and etc)
  - Risk aversion, relational contract, and liquidation value
  - Recent empirical work: Backus et al. 2020
  - **Contribution: Time-varying bargaining power**
- Tenant leasing in shopping mall
  - Affine contract form is rationalized in agency problem
    - Benjamin 1992, Brueckner 1993, and etc
  - Empirical work: Gould et al. 2005 analyze team problem in mall
  - **Contribution: Kinked contract form**

Background and Data

Shopping Mall Management and Renewal Negotiation

- A shopping mall is operated by a real estate company
  - It generates revenue through tenant rents
- Three phases of shopping mall management:
  1. Searching for new tenants
  2. Negotiating leasing contracts (including renewals)
  3. Maintaining relationships after tenants move in
- Negotiations are delegated to local managers and field staff
  - The company assigns a *manager* to each shopping mall
  - The manager assigns a *representative* to each tenant
- Typical flow of a renewal negotiation
  1. The shopping mall's initial offer: termination or a renewal proposal
  2. If termination is offered, this decision is non-negotiable
  3. Otherwise, negotiations on the terms of the new lease commence
  4. The tenant begins operations under the new leasing contract

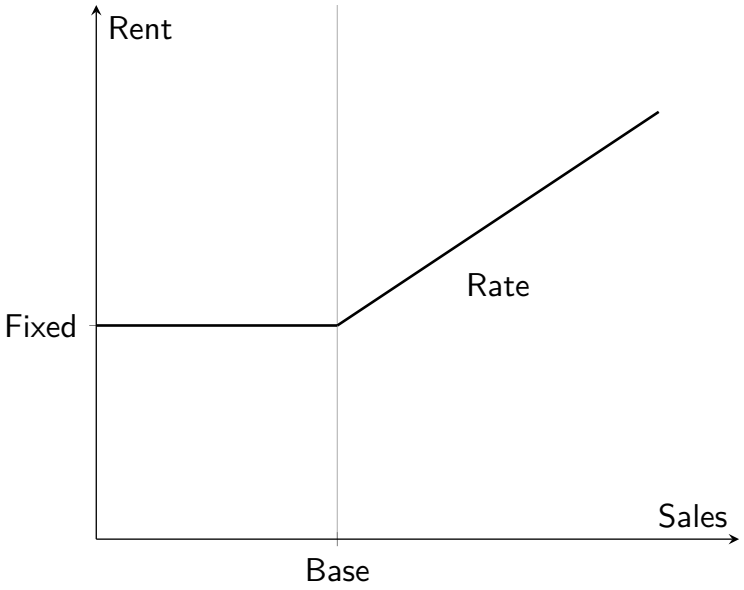
Background of Contract Negotiation

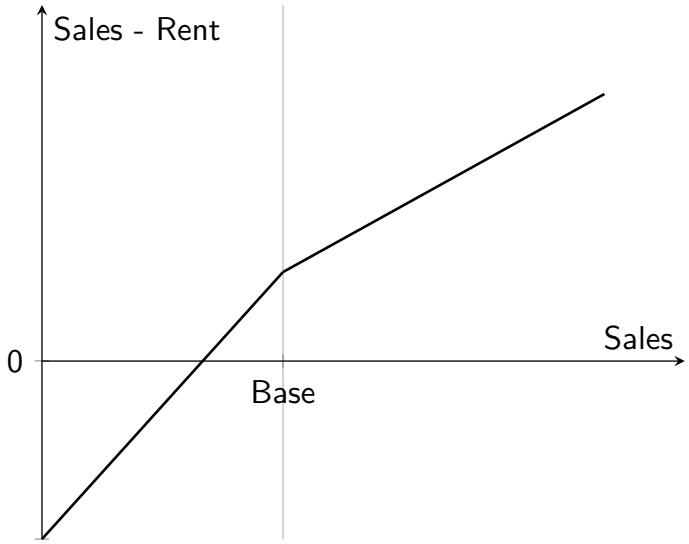


- Mall and Tenant has a negotiation table for a new contract term
  - The main issue is **rent structure**
- Each contract negotiation is composed of two tasks
  - **Earnings estimate**: Derive the sales distribution for the next period
  - **Risk sharing**: Shift the risk involved in the estimate toward each other
- Earnings estimate is not an observed average sales: professionals say
  - Tenant tries to increase the earnings estimate / Mall sticks to the realized sales history
  - Also on the bargaining table

Rent Structure

- Monthly rent is typically determined by a variant of share contract
  - Defined by parameters: *Fixed*, *Base*, *Rate*
  - Base = threshold where commission rate changes
  - Rate = commission rate after Base
- Fixed is hard to negotiate → Base plays a role in searching for a compromise





- General form of contract allows multiple kinks Go to all contracts

Data

- Data covers two shopping malls over 6 years (2017-2023)
  - Managed by same company, by different managers
  - Located in different commercial region in Japan
  - One in downtown (*Mall 0*) $\rightarrow$ More *privileged* mall
  - The other in suburb (*Mall 1*) $\rightarrow$ Less *privileged* mall
- For them, I have the following two data sources, linked as a panel data
  1. performance data
  2. contract data
- They are used for actual leasing operation of the management company
- I focus on contract renewals not contracts with new entrants

Descriptive Stats

- 226 tenants
- They operate under 443 contracts: Avg. # of renewals is 1.00
- Avg. length of lease duration is 1830 days
- Contract terms vary a lot: All monetary values are scaled by $1,000$ JPY

	Mean	Std	Min	25%	50%	75%	Max
Fixed per Area	5.26	4.11	0.15	2.74	4.51	6.05	27.69
Base per Area	62.02	43.25	3.22	42.36	54.78	60.50	435.57
Rate (%)	0.91	0.30	0.20	0.80	1.00	1.00	3.50

- Commission part is non-negligible

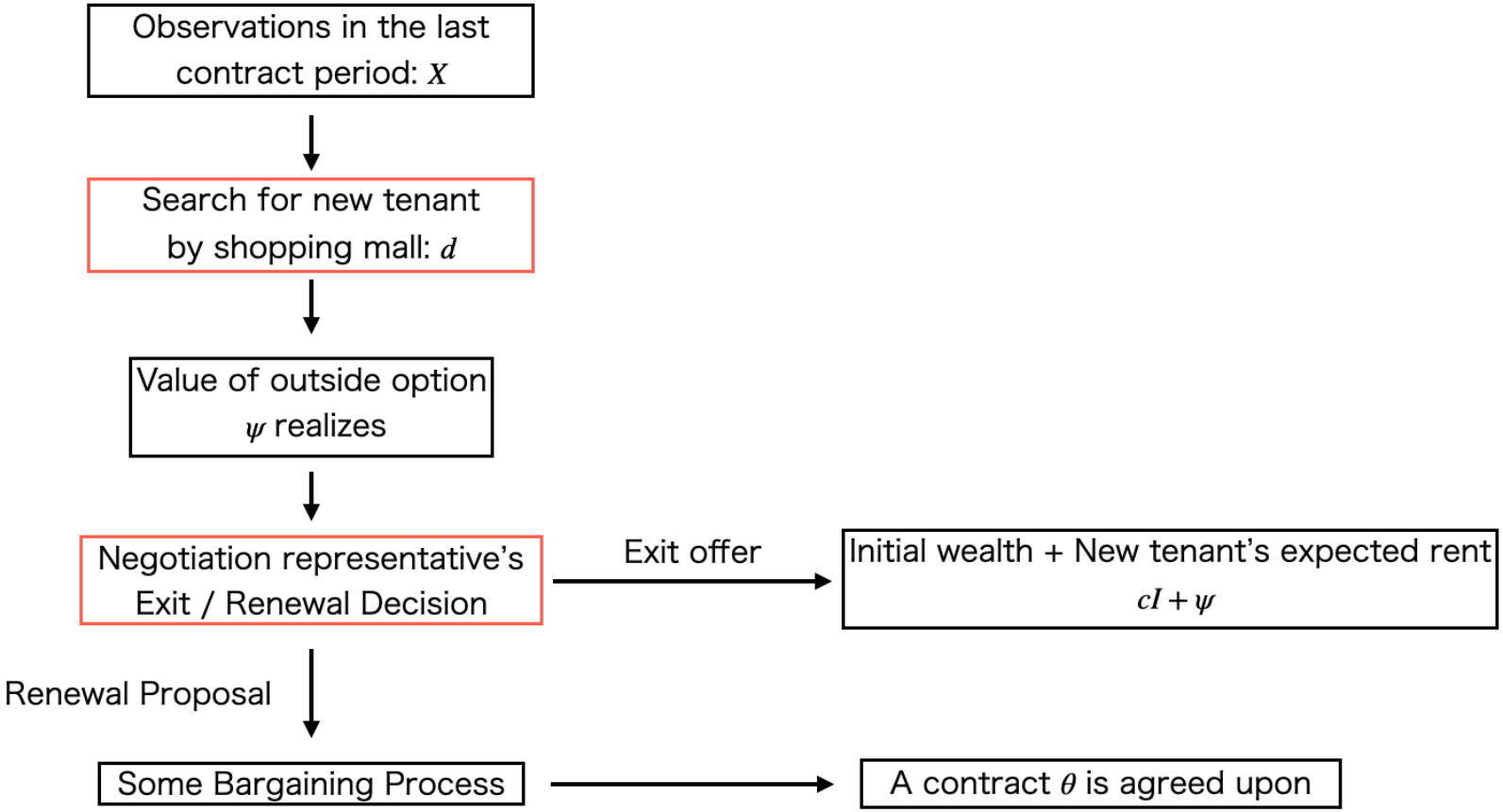
	Mean	Std	Min	25%	50%	75%	Max
Avg. Variable Rent / Total Rent	0.19	0.19	0.00	0.02	0.14	0.31	0.89
Prob. of Sales over Base	0.55	0.36	0.00	0.17	0.67	0.88	1.00

Motivating Observations and Anecdotes

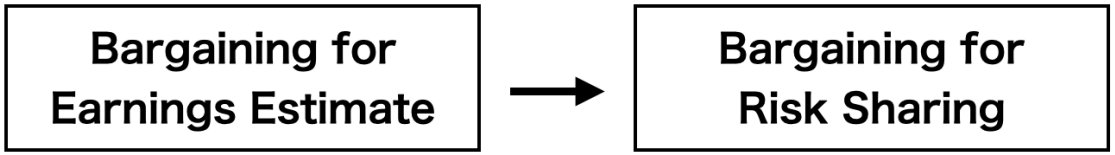
- Risk attitude matters figure
  - "Risk sharing is the primary reason to set a share contract."
  - Clearly trading off the risk (variance of rent) against the volume of rent
  - I consider both tenant and mall is risk averse with different risk aversion
- Little concern for moral hazard
  - "It's hard to imagine that sales efforts would change based on contract"
    - **Tenant side**: Shop manager is evaluated by sales not rent
    - **Mall side**: Sales-enhancing efforts (e.g., cleaning) are funded separately through charges collected from tenants
  - No bunching at Base where the commission rate starts
  - No statistically significant effect of any contract terms on realized sales table

Model

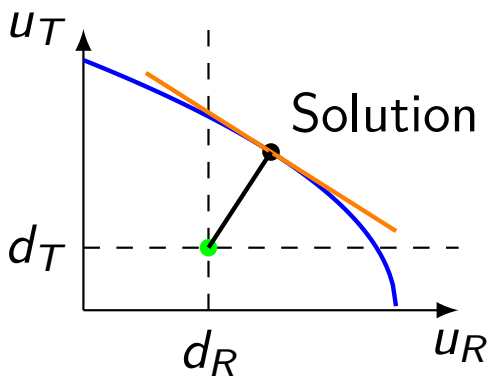
Model Overview



Model of Contract Negotiation



- The two tasks are sequentially resolved through separate Nash bargaining
- **Nash bargaining solution**
  - Primitives: Bargaining frontier, Break-up point, Bargaining power ratio
  - Outcome: Surplus split between parties



- Solution concept: **Nash-in-Nash solution**
  - Multiple interrelated Nash bargaining problems are solved under "other problems fall into the Nash bargaining solution" assumption
- Two BP ratios for the first table and the second table
  - Inverse of second BP ratio = Ratio of Risk Aversions
  - First BP ratio = *Privilege* ← Directly parametrized by covariates

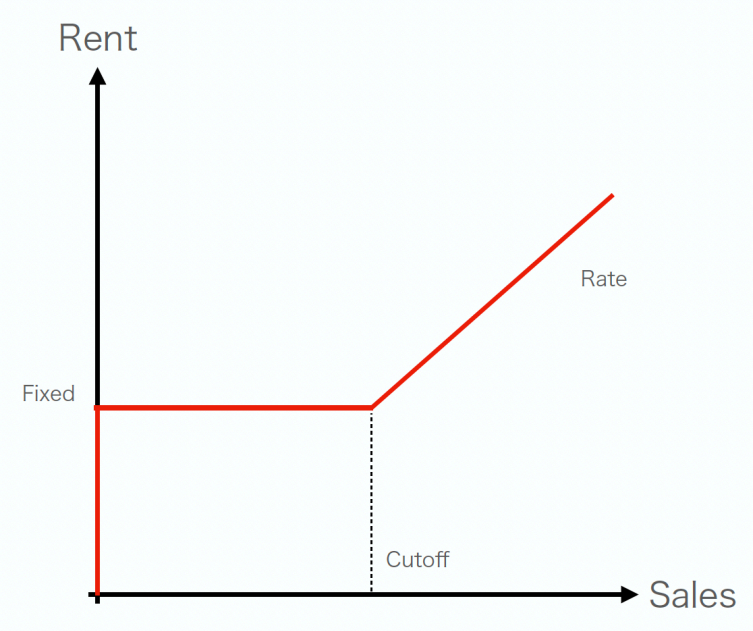
Two Bargainings

- Rent structure specifies how the sales generated in the space is divided
  - Parties face risk and uncertainty
  - Sales $S \sim N(\mu, \sigma^2)$: μ is unknown and σ^2 is known
- The two issues are sequentially bargained
  - First: *Bargaining for earnings estimate*
    - Negotiation over μ
    - Involved parties: mall manager, M, and tenant, T
  - Second: *Bargaining for risk sharing*
    - Negotiation over contract terms given μ
    - Involved parties: mall's representative, R, and tenant T
- I do not model incomplete information bargaining
  - I focus on what happens when power balance changes,
  - Not on how the balance is determined

Rent Structure

- Θ is the parameter space of mixed-type contract: $\theta = (f, b, r) \in \mathbb{R}^3_+$
  - f is Fixed, b is Base, and r is Rate
- The monthly rent is computed as follows

$$R(S; \theta) = f + r \times \max\{0, S - b\}$$



Preferences

- CRRA utility function:
  - Tenant: $U_T(S;\theta) = -e^{-\rho_T(S-R(S;\theta))}$
  - Representative: $U_R(S;\theta) = -e^{-\rho_R(I+R(S;\theta))}$
    - I is the initial wealth of the shopping mall
- Manager of the mall sticks to the realized sales information:
  - Reference point: $\hat{u}_R \equiv U_R\left(\hat{\mu};\hat{\theta}\right) + \frac{1}{2\lambda}$
    - $\hat{\mu}$ is average of realized sales in the past contract period
    - $\hat{\theta}$ is the past contract terms
    - $\frac{1}{2\lambda}$ is a bias term: Mall's average expectation for a renewal
  - Deviation from the reference point reduces the utility

Theoretical Analysis of Two-stage Sequential Bargaining

- Affine contract and mixed-type contract have the same bargaining set
  - Explicit bargaining frontier and Nash-in-Nash solution
- Byproduct: the model is *incomplete*
  - There is the set of equilibrium mixed-type contracts
  - For empirical analysis, we need to estimate *contract selection rule*
- Both R and T like higher $\mu \rightarrow$ Over optimistic contract if fully delegated
  - Optimal institutional design: the authority over μ is left to the manager
- Fundamental conflict in the bargaining for earnings estimate:
  - T likes higher μ / M dislikes too much deviation from the past realization
  - Stronger mall = Smaller μ

Continuation Decision

- Representative decides either termination or continuation
- "Expected surplus from renewal" vs "Value of outside option"

$$\mathbb{E}\left[u_R \mid \psi\right] > -e^{-\rho_R(cl+\psi)}.$$

  - ψ : monetary value of outside option
  - Expectation is taken w.r.t. B^{EE}: information gap between M and R
- Exit offer is almost non-negotiable Go to offer/outcome table
- (Omit) I also model ψ which is affected endogeneous search behavior Go to model of outside option

Empirical Strategy

Empirical Strategy: Variables

- An ongoing contract is denoted by a pair of i (tenant) and k (mall)
- Any contract is numerated with the contract number τ starting from 1
- Contract period (in months) is denoted by $T_{ik\tau}$
- Performance is measured for every months: $\tilde{X}_{ikt}$ for $t \in \{1, \cdots, T_{ik\tau}\}$
  - Main measure: *rank of sales per area within a mall*
- For an ongoing contract, we make a measure of performance:

$$X_{ik\tau} = \frac{1}{T} \sum_{t=1}^{T_{ik\tau}} \tilde{X}_{ikt}$$

  - the average of the performance measure during the contract period

Empirical Strategy: Parametrization / Estimation

- Logarithm of bargaining power ratio

$$\ln B_{ik\tau}^{EE} = X_{ik\tau}'\gamma + \varepsilon_{ik\tau}^{EE},$$

  - Three variables related with tenant; such as *rank of sales per area*
  - Four variables related with the shopping mall; such as *mall total sales*
  - $\varepsilon_{ik\tau}^{EE}$ is unobserved factor

- (Omit) Value of outside option = linear part + error term $\tilde{\varepsilon}^o$

- Two random terms are i.i.d. joint Normal: $(\epsilon^{EE}, \tilde{\varepsilon}^o) \sim N(0, \Sigma)$

- Likelihood function is constructed as in Tobit model:

$$\text{Likelihood} = \begin{cases} Pr(\text{observe contract } \theta_{ik\tau}) & \text{if continuation} \\ Pr(\text{exit}) & \text{otherwise} \end{cases}$$

- Maximize log-likelihood under the constraint of Nash bargaining solution
  - Earnings estimate is endogeneous

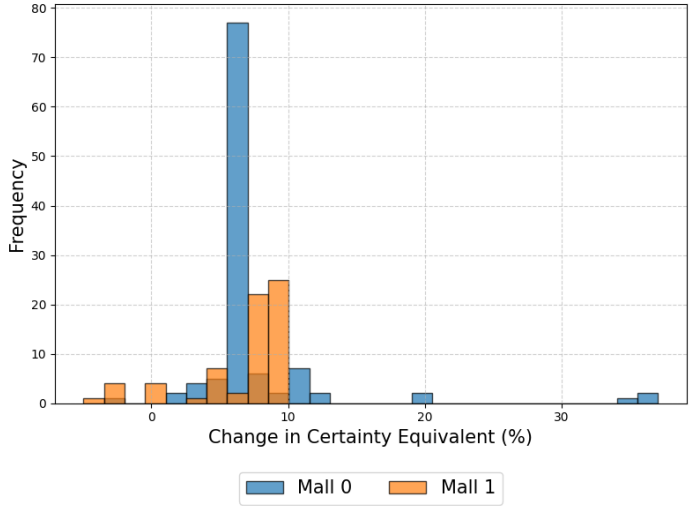
Identification

- Two sets of parameters
  1. Common across tenants within a mall
     - risk aversion of mall and marginal effects of covariates etc
  2. Contract-specific
     - Risk aversion of tenant: $\rho_{i\tau}$
     - Agreed upon expected sales (endogeneous parameter): $\mu_{ik\tau}$
- The first set is identified by the variation of exit offer
  - The second group is removed from the continuation decision
  - WHY?
    - Assumption $B^{RS} = \frac{\rho_T}{\rho_R}$ eliminates ρ_R from the decision
    - Cooperative approach allows me to parametrize $B^{EE}_{ik\tau}$ determining $\mu_{ik\tau}$
- The second set is identified by conditions of Nash solution
  - Two agents $\rightarrow$ Two surplus expressions $\rightarrow$ Two conditions

Result

What makes mall strong?

- Static characteristics
  - More number of total tenants in the mall
  - More number of total customers purchasing in the mall
  - Tenant located in the smaller retail space
- Time-varying characteristics
  - Tenant's lower rank in terms of sales per unit area within the mall
- Time-varying components are significant
  - Simulate static version of B^{EE}
  - Remove all the effects from time-varying variables from B^{EE}
  - Change in certainty equivalence relative to the actual surplus
  - About 10% varies by such variables



Inspection of Contract Selection Rule

- Empirical analysis on contract term = Estimate contract selection rule
- SUR model for the three parameters, Fixed, Base, and Rate:

$$\begin{pmatrix} Fixed \\ Base \\ Rate \end{pmatrix} \sim \text{Bargaining Powers} + \text{Covariates} + \begin{pmatrix} \epsilon_f \\ \epsilon_b \\ \epsilon_r \end{pmatrix}$$

Empirical Contract Pattern

Mall	Mall 0			Mall 1		
Dep. Var	Rate	Base	Fixed	Rate	Base	Fixed
Avg. Sales	0.00204***	-0.135	0.00922	0.00533***	0.461***	0.0274
	(0.000720)	(0.0843)	(0.0134)	(0.000962)	(0.121)	(0.0202)
Var. Sales	0.000881	0.671***	0.0552**	-0.00307	-0.130	0.0299
	(0.00116)	(0.136)	(0.0216)	(0.00321)	(0.404)	(0.0675)
Privilege	-0.0480***	-2.987*	-0.0569	-0.0400***	-3.086**	-0.874***
	(0.0136)	(1.590)	(0.253)	(0.0101)	(1.275)	(0.213)
Risk loving	0.125***	-2.154	0.341	0.0985***	-1.350	-0.157
	(0.0121)	(1.416)	(0.225)	(0.00789)	(0.994)	(0.166)
N	90	90	90	62	62	62

- Greater privilege $\rightarrow$ Lower earnings estimate $\rightarrow$ Smaller fixed and lower base
- Risk loving $\rightarrow$ Higher rate
- Contract selection rule itself is consistent with risk attitude
  - Mall 0 (Risk averse): Large volatility $\rightarrow$ Higher Fixed
  - Mall 1 (Risk loving): Higher sales $\rightarrow$ Higher base and Higher rate

Concern: Endogeneous Risk Adjustment

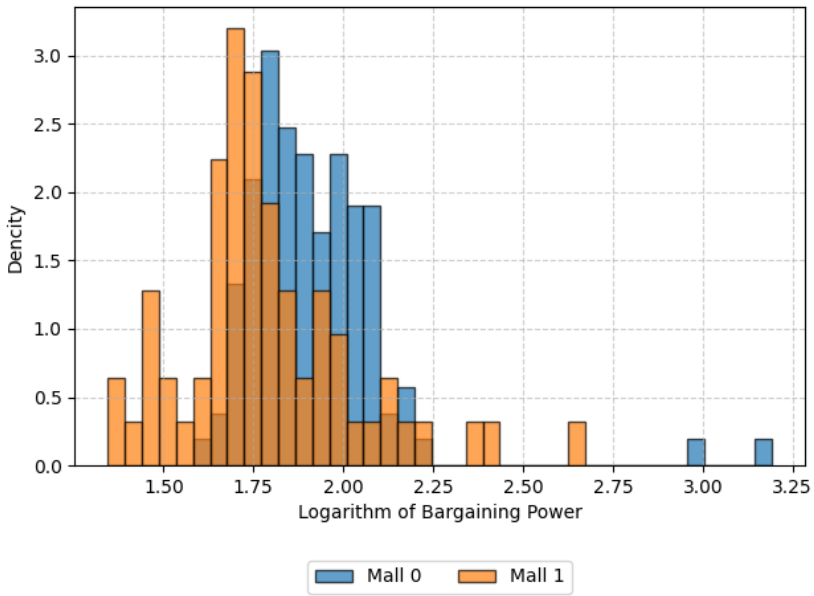


Figure: Privilege

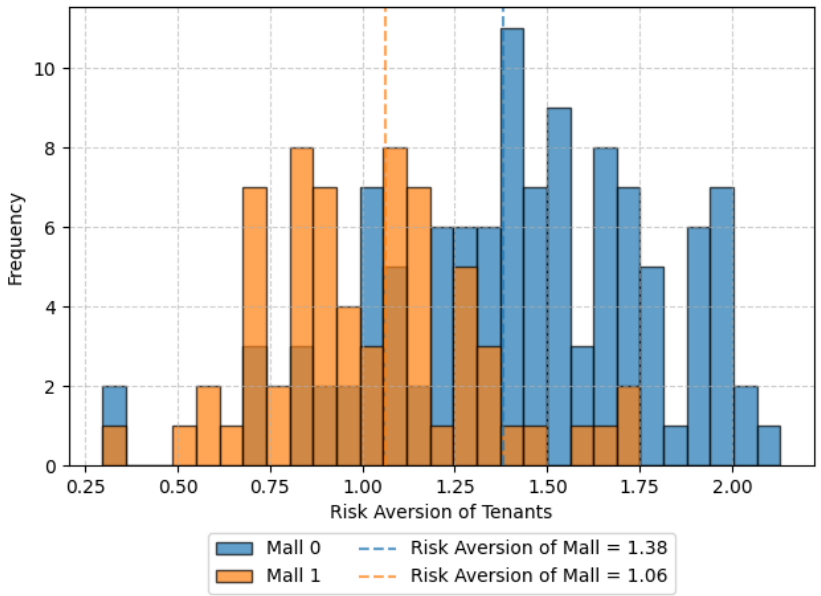


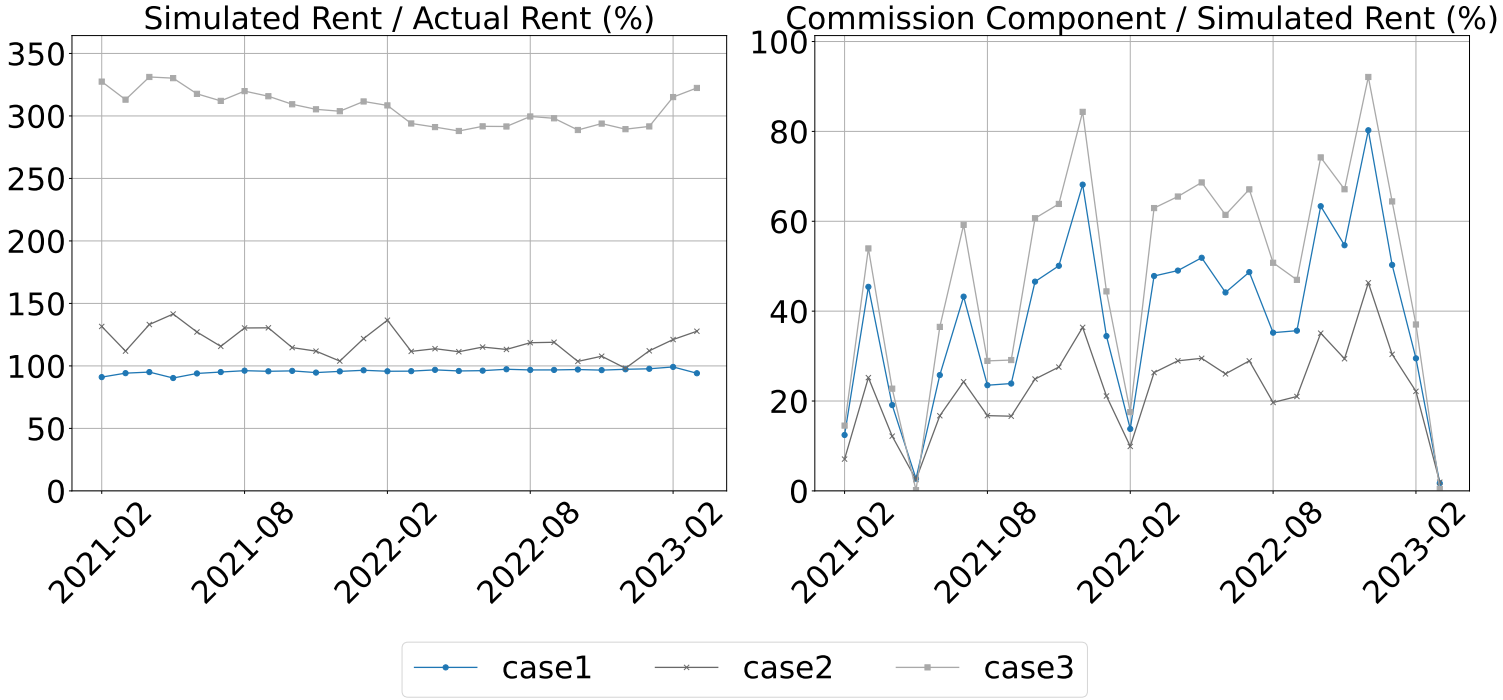
Figure: Risk aversion

- Mall 0 : Superior position / Lower earnings estimate / Greater risk aversion
- **Lower earnings estimate ↔ Greater risk aversion**
  - Risk averse manager is assigned to such mall
  - No need to take risks in superior position

Simulation Setting: Fairer Bargaining

- Question: How will the transfer change in the fairer situation?
  - Why is this an empirical question? $\rightarrow$ Two paths exist
    1. Weaker positions $\rightarrow$ Higher earnings estimate $\rightarrow$ Larger fixed rent
    2. Weaker position $\rightarrow$ Less risk averse $\rightarrow$ More commission component
- Possible situation:
  - Mall 0 potentially abuses its privilege when making contracts
  - Regulator enforces the fairer bargainings in renewal through warnings
- Scenarios:
  - Case 1: Replicate actual rents
  - Case 2: Mall 0's privilege is determined in the same way as in Mall 1
  - Case 3: Case 2 + Mall 0's risk aversion is set to the same value of Mall 1

Result: Not Always Less Rent in Fairer Bargaining



- Case 2: weaker mall with fixed risk attitude
  - Drop in commission component; consistent with second mechanism
  - Increase in total rent: actual base may be too high
- Case 3: weaker mall becomes risk-loving
  - Rise in commission component: first mechanism dominates
  - Large increase in total rent revenue

Conclusion

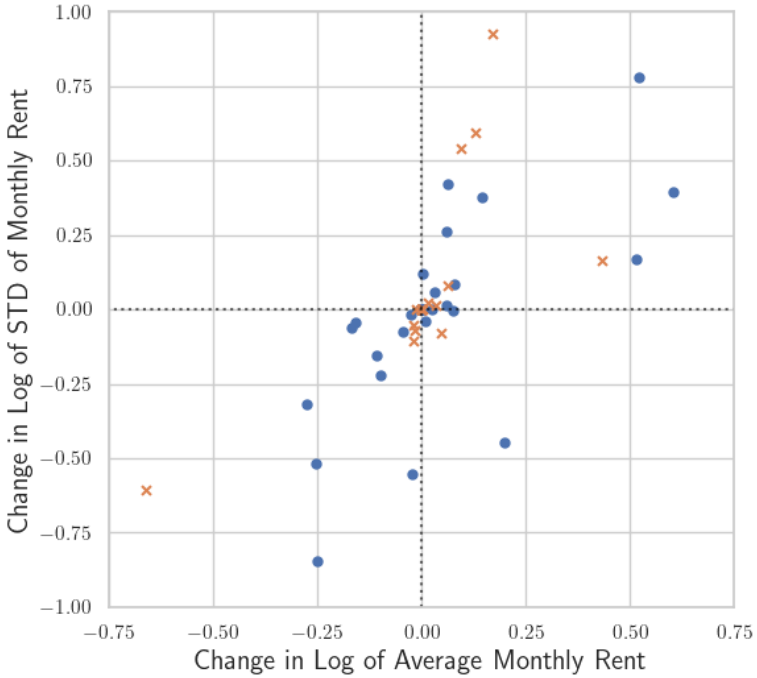
- I analyze tenant contracts in shopping malls empirically
- I find that fair bargaining has a complex influence on contract terms
  - A weaker-positioned mall may adopt riskier contract terms
- Regulators must be cautious when intervening in contract negotiations
  - Such interventions may lead to unexpected changes in contract terms
  - ... and the following transfers among them
- Discussion
  - Collective bargaining (negative / positive spillovers)
    $\rightarrow$ Nash-in-Nash can be applied but more data is necessary
  - Agency issue
    $\rightarrow$ Several studies consider bargaining over contract under agency issue
  - Private information

Appendix

Obs. : Risk Attitude Matters

back to slide

- For all contracts:
  - Average monthly rent
  - Standard deviation of monthly rents
- Changes in these metrics over two successive contracts are plotted
- Higher rent must be accompanied by larger variance



Moral Hazard?

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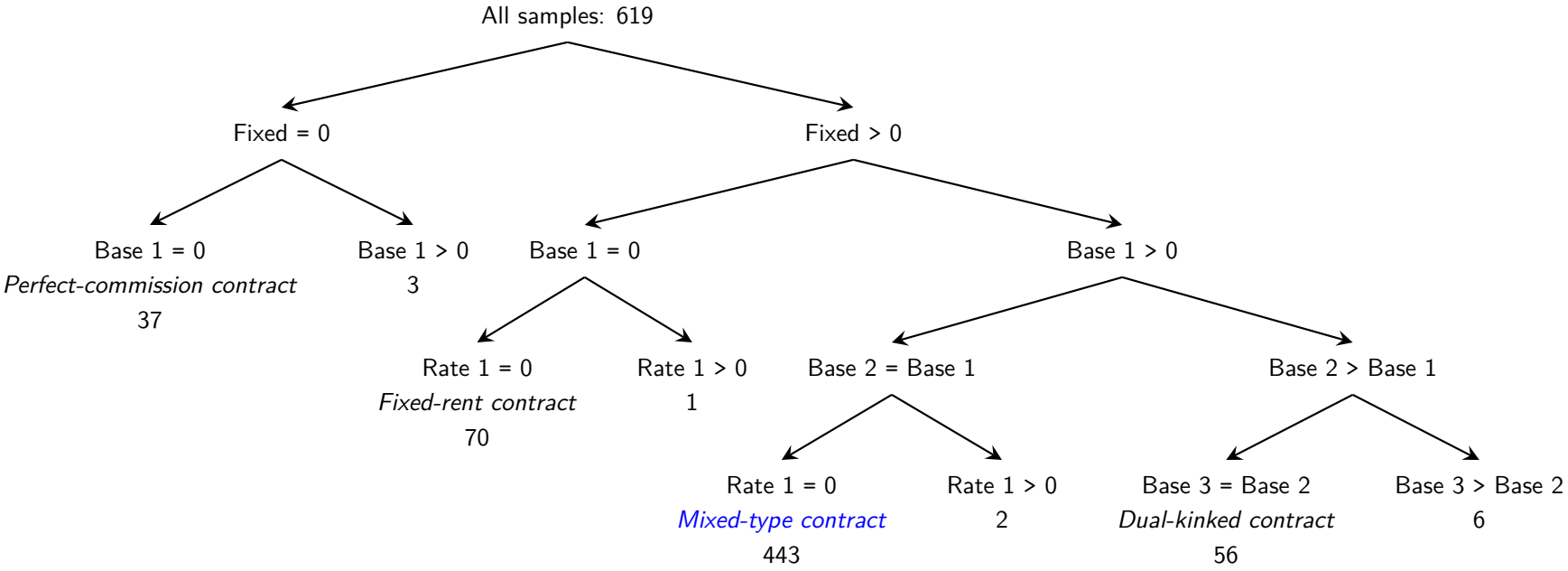
Dep. Var. : Sales	(1) Level	(2) Diff	(3) Diff / Change
Rate	−0.063	−0.049	−0.180
	(0.219)	(0.222)	(0.374)
Base	0.069	0.000	0.000
	(0.206)	(0.000)	(0.000)
Fixed	2.839**	0.000	0.000
	(1.319)	(0.000)	(0.000)
Observations	197	197	156
Adjusted R^2	0.992	0.964	0.967

- Moral hazard test: Do contract terms affect tenant performance?
  - Regress average monthly sales on contract terms and controls
  - No evidence that higher commission rates reduce tenant effort
- Institutional features mitigate moral hazard: mystery shopper surveys assess

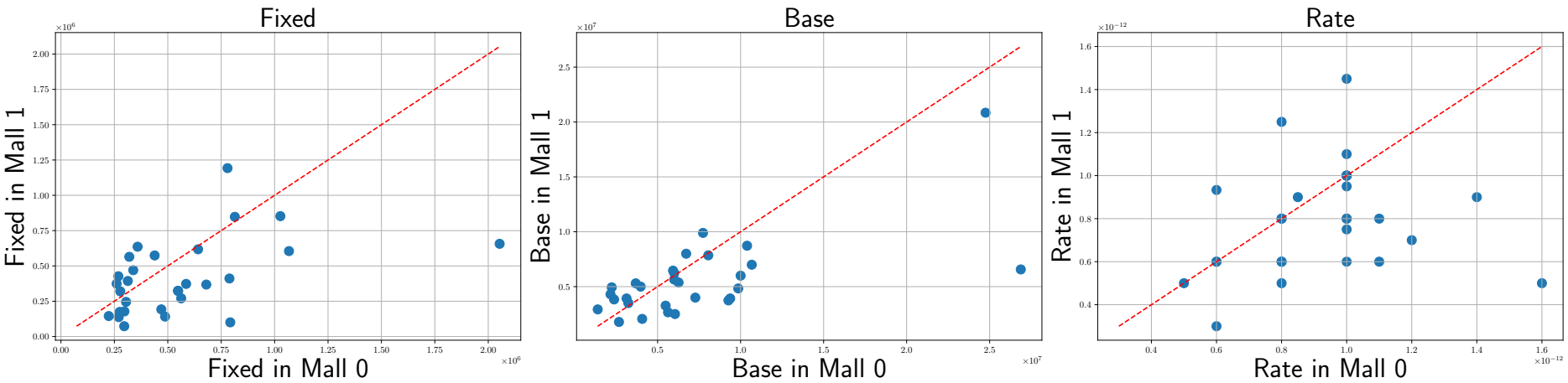
All Rent Structures

back to mixed type contract



Common Brands

back to descriptive stats



Contract Type	1	2	3	5	6	7	8
Mall 0	2	0	10	58	2	1	2
Mall 1	7	1	11	44	0	7	0

Table: Distribution of Rent Structures of Common Brands by Malls

Exit Offer

back to model

Initial Offer \ Next Contract	**exit**	**same**	**up**	**down**	**updown**	**TOTAL**
exit	52	3	1	2	2	60
same	15	137	11	13	13	189
up	6	60	49	5	6	126
down	5	6	5	11	5	32
updown	1	1	3	5	9	19
TOTAL	79	207	69	36	35	426

Table: Initial Offer and Resulting Contract

Note: The table presents the relationship between the first initial offer (rows) and the resulting contract type (columns). Both the initial offer and the resulting contracts are categorized into five groups: exit, same, up, down, and updown. The numbers in the cells indicate the count of contracts corresponding to each combination of initial offer and resulting contract.

Value of Outside Option

back to model

- Search intensity (d): the count of the meeting with potential tenants
- d affects ψ

$$\psi = \tilde{\psi} + \beta_o d$$

  $\tilde{\psi}$ is the baseline value of outside option

- Endogeneity: Better outside option might decrease the search intensity
  - $\tilde{\psi}$ and d are simultaneously determined
  - No full model, just a correlation structure $\rightarrow$ *Control function approach*

Parametrization: Value of Outside Option

- ψ basically captures the market demand for the retail space
- ψ is determined by three parts:
  - public information regarding the retail space, Z
  - search behavior conducted by the shopping mall, d
  - disturbance capturing the market's unobserved demand, $\tilde{\epsilon}^o$
- W : IV for search intensity $\rightarrow$ tenant specific performance measure
- Linear model (**??**):

$$
\begin{aligned}
\psi_{ik\tau} &= \left(Z'_{ik\tau}\gamma_o^{\psi} + \varepsilon_{ik\tau}^{o}\right) + \beta_o d_{ik\tau} \\
d_{ik\tau} &= Z'_{ik\tau}\gamma_o^{d} + W'_{ik\tau}\delta + \nu_{ik\tau}^{o} \\
\varepsilon_{ik\tau} &= \kappa\nu_{ik\tau}^{o} + \tilde{\varepsilon}_{ik\tau}^{o}
\end{aligned}
$$

  - $\tilde{\varepsilon}_{ik\tau}^{o}$ is an exogeneous shock to the value of outside option